

Samuel Ferreira

Software Development Student | Back-end and Full Stack

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PROFILE

Systems Analysis and Development student building back-end services and full-stack applications around real business workflows. Practical experience with API design, relational databases, authentication, transactional rules, automated tests, Docker, and Windows development tooling. Looking for an internship or junior software development role.

SELECTED PROJECT

ZirphasOS - Repair Shop Management | Python, FastAPI, PostgreSQL, SQLAlchemy, Docker

A repair-order API that connects customer intake, technician work, estimates, inventory, payments, and delivery through controlled state transitions.

- Implemented JWT authentication, role-based access, Argon2 password hashing, and private customer tracking tokens stored as hashes.
- Modeled versioned estimates, transactional part consumption, idempotent payments, and delivery after full payment.
- Documented and automated 68 tests, including PostgreSQL concurrency and financial-integrity scenarios.

ADDITIONAL ENGINEERING WORK

- Built a React business dashboard with derived metrics, inventory alerts, and a responsive production build.
- Created a Node.js and Discord.js support-ticket workflow with permissions, private channels, logs, and interactive components.
- Developed a Python and SQLite CLI for products, inventory, sales, category reports, and automated tests.
- Implemented small security and automation tools for repository checks, structured data, and repeatable workflows.

TECHNICAL SKILLS

Back end	Python, FastAPI, Go, Node.js, REST APIs, JWT, RBAC
Front end	React, TypeScript, JavaScript, HTML, CSS, responsive UI
Data	PostgreSQL, SQLite, SQLAlchemy, Alembic, SQL, transactions
Delivery	Git, GitHub Actions, Docker, Compose, Windows, Linux containers
Quality	Pytest, Go tests, integration tests, concurrency tests, documentation

EDUCATION

Systems Analysis and Development - In progress

Current focus: software architecture, databases, data structures, operating systems, networks, requirements, and software engineering practices.

Public repositories include setup instructions, documented limitations, and validation records so that project claims can be checked against the code.